

Agent-swarm physics: hypothesis compendium

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Two RevTeX pages per hypothesis follow; click a row to jump to it. Ratings are estimates calibrated across hypotheses by the coordinator (rubric: [writeup/hypothesis-pages/RUBRIC.md](#)). **Completion:** estimated progress of the research direction (idea 5%, round 1 done 35–50%, holdout run 55–65%, robustness and causal designs 70–85%, settled 90–100%).      **Faithfulness** (0–5, scoped): how well the model holds for what it claims (3 = descriptive or primary holdout passed; 4 = supported; 5 = faithful and mechanistic). **Usefulness** (0–5, unscoped): what an operator or alignment researcher can do with it, true or not (1 = vocabulary only; 2 = a diagnostic; 3 = a validated monitor or design rule; 4 = a steering lever; 5 = a transferable control knob). 

     

ID	Hypothesis	Complete	Faithful	Useful	p.
H01	Emergent superagents exist Rooms are more like-minded than random groups but missed both locked sizes; order looks like room fields				4
H02	Inferred couplings reflect real influence Activity-timing Ising couplings are null-level and missed the installed #45 leader; only shared lulls remain				6
H03	Swarm activity is self-exciting, and its criticality is set by the go... Subcritical everywhere (n about 0.2-0.4), no coupling-mode effect; only fast cross-triggering beats nulls				8
H04	External forcing reshapes the response kernel, reversibly Nudges act only on their target, after minutes of delay; the hours-reversal holdout was falsified in reverse				10
H05	Cutting the cross-room channel lowers entropy production, and rooms b... Rooms cut chat coupling (holdout DiD passed; verdict inconclusive); work isn't room-bound; EP unmeasurable				12
H06	Free weeks show neutral cooperative dynamics with a cooperator core (draft) specified (shortlist S6 / HH42; approved by Vivian 2026-10-03; started 2026-10-04). Exploratory round 1 starting; pre...	—	—	—	14
H07	The RPG forks diverge from a common ancestor at a measurable rate Forks diverge gradually in code, in team-chosen jumps in content; copying dominates; no leakage; n = 2				16
H08	Context is the coupling Responses wait for the next model call (10/11 periods); nudges lag read-out ~3 min; erasure cuts coupling 18%				18
H09	Agent swarms have an effective thermodynamics Thermo framing adds little; idle agents are timer-gated (only @-mentions get through); memory is a set point				20
H10	Goals are Legendre pushes: a swarm's response to an assigned goal is ... Goals aren't Legendre pushes: free-week fluctuations don't predict who follows; agents converge instead				22
H11	Division of labor vs herding is the sign of a Potts coupling, set by ... Agents herd onto shared projects in 11/14 weeks whatever the goal; spread appears only as fixed ownership.				24
H12	Grouptthink is dimensional collapse: few collective modes, shrinking p... One market mode, no kickoff collapse; low-dimensional weeks are agents looping on their own text.				26

ID	Hypothesis	Complete	Faithful	Useful	p.
H13	Model families carry their own fields, and couple by family Model families share a stable content field that is writing style; no family homophily; rooms carry coupling.	45%	1.5	2	28
H14	Entropy production of behavior-state sequences: per-family arrows of ... Irreversibility lives in fine tool actions and scaffold steps; no per-family (HH19) or collective (HH67) arrow	45%	1.5	1.5	30
H15	Semantic information through natural scrambles: which information kee... Losing memory costs nothing visible in a day; wiping the context window costs ~45% of output for 10 turns	40%	1.5	2.5	32
H16	Agents get stuck in metastable traps and escape by Kramers kinetics; ... Agent traps age, not Kramers-escape; one directed message at a pause gate helps, more add little.	42%	1.5	2.5	34
H17	Behavior is a Markov state model with a few metastable sets; its mixi... Action-class behavior is non-Markov; slow sets are GUI/shell modes and idling; t2* doesn't flag stuck periods	40%	1	1.5	36
H18	Attention dilutes as 1/k: response to a message falls with the number... Attention dilutes as ~k ^{-0.6} in all 16 periods (not 1/k): per-pair engagement falls with room size	42%	2	2	38
H19	One curve for all periods: per-period loop gains collapse onto a func... No common loop-gain curve: activity and talk channels move oppositely; Hawkes-to-Ising mapping holds.	40%	1	1.5	40
H20	Aging: day-to-day content autocorrelation depends on time since kicko... No aging: swarm content settles within ~4 days of a kickoff, then decorrelates at an age-independent rate	40%	1	1.5	42
H21	The debate week (#12) is a two-sublattice antiferromagnet Debate teams leave no two-sublattice order in content; only a faint stance tilt that reverses after verdicts	40%	1	1.5	44
H22	Private, conflicting goals make #51 a spin glass #51 is no spin glass: roles pin positions, couplings are mostly positive, rivals co-move, no collective states	40%	1	1.5	46
H23	Distilling the village into a fine-tuned leader copies its vocabulary... Distilled leader kept a trace of its corpus style but none of its plans; its corpus was Kimi talking to itself	40%	1.5	1.5	48
H24	In the forecast week (#21), switching from independent drafts to comp... Forecast week: the coupling switch was real but early; alignment didn't jump and herding was an artifact	40%	1	1.5	50

ID	Hypothesis	Complete	Faithful	Useful	p.
H25	A distance-to-criticality dial, T/T _c , per window and per channel (draft) running. Exploratory round 1 in progress (predictions written 2026-10-04 01:45 UTC, before any dial was computed on r...	—	—	—	52
H26	The swarm is near-critical in what it says but subcritical in when it... (draft) running (round 1). Promoted 2026-10-04 by Vivian from HH108 (usefulness-first batch, wave 2). Observables, nulls and ...	—	—	—	54
H27	Critical slowing down warns of herding waves hours ahead (draft) specified (promoted 2026-10-04 by Vivian from HH109). Exploratory round 1 running: predictions, observables and nulls...	—	—	—	56
H28	Links are the contagion vector of herding (draft) exploratory round 1 in progress (2026-10-04). Predictions below were written before any real-data run.	—	—	—	58
H29	Driver nodes: where an operator message moves the whole swarm (draft) exploratory round 1 running (2026-10-04). Observables, nulls and predictions written 2026-10-04 01:45 UTC, before an...	—	—	—	60
H30	An operator-susceptibility gauge χ_{op} : how steerable is the swarm tod... (draft) exploratory round 1 running (2026-10-03). Predictions and nulls written below before any real-data H30 statistic; syn...	—	—	—	62
H31	Consensus time scales with the interaction graph's spectral gap (draft) exploratory round 1 running (2026-10-03). Predictions below were written 2026-10-03, before any real-data run.	—	—	—	64
H32	A net information current identifies de facto leaders (draft) exploratory round 1 running (2026-10-03). Pre-registration below written before any real-data outcome.	—	—	—	66
H33	The diversity–productivity curve is an inverted U: the swarm's operat... No diversity-productivity optimum: agent-day content PR carries no usable information about write output (flat above ...	40%	0.5	1.5	68
H34	Idea cascades follow a power law whose exponent reads off the distanc... (draft) running. Exploratory round 1 started 2026-10-04 01:30 UTC; predictions below were written before any real-data run.	—	—	—	70
H35	The nudger is a measurably inefficient Maxwell demon (draft) exploratory round 1 running (2026-10-03). Observables, nulls and dated predictions written below before any real-data...	—	—	—	72
H36	A reorganization alarm: susceptibility and multi-information peak at ... (draft) exploratory round 1 running (agent, 2026-10-04). Observables, nulls, the alarm rule and the predictions below were wr...	—	—	—	74
ID	Hypothesis	Complete	Faithful	Useful	p.
H37	Conflict lives in stance, not topic: stance spins are antiferromagnetic (draft) exploratory round 1 running (2026-10-04). Observables, nulls and predictions written 2026-10-04 01:35 UTC, before any...	—	—	—	76
H38	Joint silences are platform stalls (draft) specified (promoted 2026-10-04 by Vivian from HH94). Exploratory round 1 running. Predictions written 2026-10-04 01:2...	—	—	—	78
H39	Catalysts vs. fields (draft) round 1 running (2026-10-04, UTC). Design, observables, nulls and predictions below were written before any H39 stati...	—	—	—	80